CHE 201 Fall 2019 Quiz 3

Duc Huy Nguyen

TOTAL POINTS

85 / 100

QUESTION 1

1 1 10 / 15

- **- 0 pts** Correct
- ✓ **- 5 pts first one is wrong**
- **- 5 pts** second one is wrong
- **- 5 pts** third one is wrong

QUESTION 2

2 2 75 / 85

- **- 0 pts** Correct
- **- 15 pts** mass balance is incorrect
- **- 20 pts** energy balance is incorrect
- **- 10 pts** h1 is incorrect
- ✓ **- 10 pts h2 is incorrect**
- **- 10 pts** h3 is incorrect
- **- 15 pts** m3 is wrong

CHE 201 INTRO TO THERMODYNAMICS **November 13th, 2019**

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QUIZ 3: (100 pts) (Close book/closed notes) Show all your work for full credit

1.(15 points) State whether the following statements are **True of False (5 pts each)**

False A thermodynamic cycle cannot convert heat completely into work.

False The change in entropy (ΔS) for a system cannot be less than zero.
universe

False A closed system process that violates the 2nd law of thermodynamics must also violate the 1st law of thermodynamics.

2. (85 points) A stream of superheated steam at 3 bar and 280 °C with a mass flow rate of 400 kg/s (stream 1) is mixed with a stream of liquid water at 3 bar and 20 °C (stream 2) in an adiabatic mixing chamber. The resulting mixture exits as a saturated vapor at 3 bar (stream 3). Assume steady state operation and neglect any changes in kinetic and potential energy. Calculate the mass flow rate of the liquid water (stream 2) entering the mixing chamber, in kg/s.

$$\frac{dE_{cv}}{dt} = \dot{Q}_{cv} - \dot{W}_{cv} + \sum_i \dot{m}_i\left(h_i + \frac{V_i^2}{2} + gz_i\right) - \sum_e \dot{m}_e\left(h_e + \frac{V_e^2}{2} + gz_e\right) \qquad \dot{m} = \frac{AV}{v}$$

1 joule= 1 N.m 1 N= 1 kg.m/s²

Schematic

Model: - Control volume - Adiabatic → Q=0 - PE=KE=0
- Steady state - 2 inlets, 1 outlet
- $\Delta V = 0 \Rightarrow W = 0$

Analysis:

Mass balance: $\dot{m}_1 + \dot{m}_2 - \dot{m}_3 = \cancel{\frac{dm_{cv}}{dt}}^{0}$

$\Rightarrow \dot{m}_1 + \dot{m}_2 = \dot{m}_3$

$\Rightarrow \dot{m}_3 = (400 \text{ kg/s}) + \dot{m}_2$

Energy balance:

$\cancel{\frac{dE_{cv}}{dt}}^{0} = \cancel{\dot{Q}_{cv}}^{0} - \cancel{\dot{W}_{cv}}^{0} + \dot{m}_1\left(h_1 + \cancel{\frac{V_1^2}{2}}^{0} + \cancel{gz_1}^{0}\right) + \dot{m}_2\left(h_2 + \cancel{\frac{V_2^2}{2}}^{0} + \cancel{gz_2}^{0}\right) - \dot{m}_3\left(h_3 + \cancel{\frac{V_3^2}{2}}^{0} + \cancel{gz_3}^{0}\right)$

$\Rightarrow 0 = \dot{m}_1 h_1 + \dot{m}_2 h_2 - \dot{m}_3 h_3$

Table A-4, at 280°C, 3bar, $h_1 = 3028.6$ KJ/kg

Table A-3, at 3bar, 20°C, $h_2 = h_f = 561.47$ KJ/kg

Table A-3, 3 bar, $h_3 = h_g = 2725.3$ KJ/kg

$\Rightarrow 0 = (400 \text{ kg/s})(3028.6 \text{ KJ/kg}) + \dot{m}_2(561.47 \text{ KJ/kg}) - (\dot{m}_2 + 400 \text{ kg/s})(2725.3 \text{ KJ/kg})$

$\Rightarrow \dot{m}_2 = 56.1$ Kg/s